DO NOW — RECALL

Boolean logic, gates and truth tables

Name: _____ Class: _____ Date: _____

1. What is the difference between a stack and a queue when removing items?

2. What must be true about an array before you use binary search?

3. For int[] numbers = {5, 9, 14, 18, 23}; , what value is stored at zero-based index 3?

4. A binary search looks for 23 in the sorted array [2, 7, 11, 16, 23, 31, 44] . Which zero-based indexes are checked, in order?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

A stack removes the most recently added item (LIFO); a queue removes the earliest added item (FIFO).

2. Model answer

The values must be sorted in the order expected by the search.

3. Model answer

18. C# arrays start at index 0.

4. Model answer

Indexes 3 → 5 → 4 (values 16 → 31 → 23). The search narrows the sorted range each time.

By the end, you can…

• translate a problem into AND, OR, NOT and XOR logic

• convert between Boolean expressions and gate diagrams

• complete truth tables systematically

Vocabulary — know these terms

Recall the term — Boolean

Recall the term — AND

Recall the term — OR

Recall the term — NOT

Recall the term — XOR

Recall the term — logic gate

Recall the term — truth table

Recall the term — expression

New Learning · Boolean logic gates

Original explanation and illustration · part 1 of 3

From switches to logic gates

Imagine two switches, A and B , controlling a light, Q .

Learn what each gate does and looks like . Only then write expressions and combine gates.

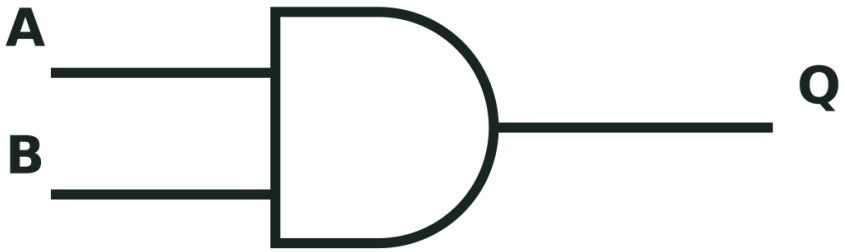
1 = ON / TRUE; 0 = OFF / FALSE; A, B = inputs; Q = output

Read the diagram from left (inputs) to right (output). A gate applies one rule.

01 · AND — both inputs must be 1

A gate takes an input (or inputs) and produces one output, Q . For AND, both inputs must be 1.

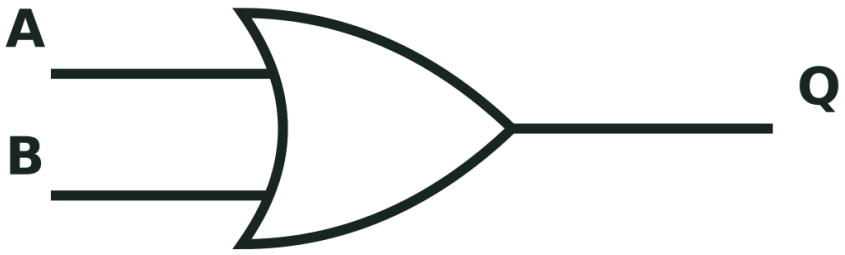
Example: 1 AND 0 = 0; 1 AND 1 = 1.



02 · OR — at least one input is 1

Either input can turn Q on. Both can be on too.

Example: 1 OR 0 = 1; 1 OR 1 = 1.



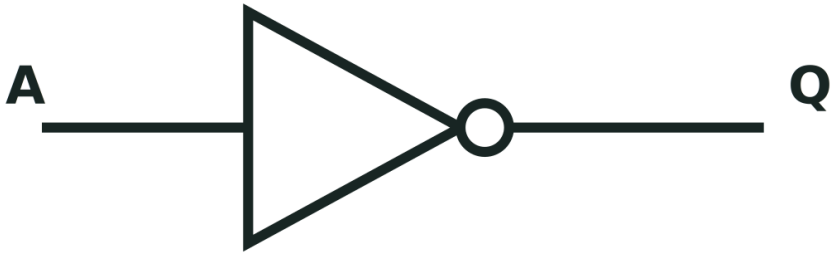
New Learning · Boolean logic gates

Original explanation and illustration · part 2 of 3

03 · NOT — reverse one input

NOT has just one input . The small circle on the output is the inversion symbol.

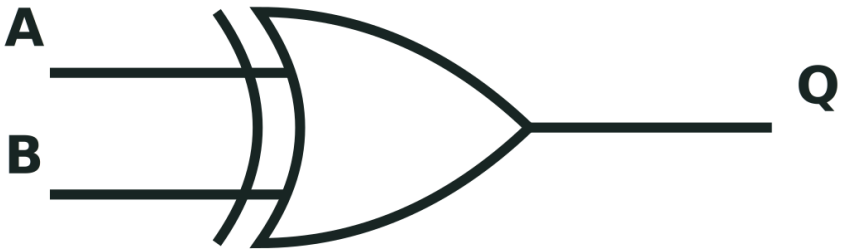
Example: NOT 1 = 0; NOT 0 = 1.



04 · XOR (⊻) — exactly one input is 1

XOR is different from OR: when both inputs are 1, XOR gives 0. Notice the extra curved line at the input.

Symbol: Q = A ⊻ B. Compare: 1 ∨ 1 = 1, but 1 ⊻ 1 = 0.



05 · Words first, then Boolean symbols

These symbols are shorthand for the rules you have just seen:

Write XOR using ⊻ : Q = A ⊻ B (1 only when A and B differ). Unlike OR (∨), XOR excludes the case where both inputs are 1. A, B and Q name inputs and output, not gates.

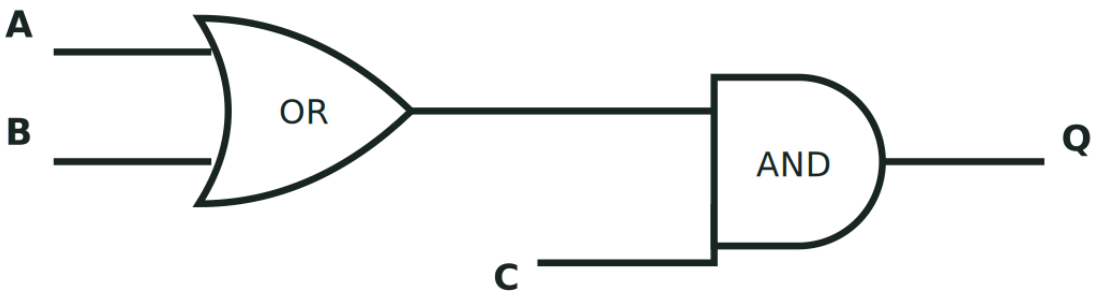
New Learning · Boolean logic gates

Original explanation and illustration · part 3 of 3

06 · Connect two gates in order

Say the logic aloud: A OR B, then AND C.

Brackets mean the OR gate is worked out before its output goes into AND.



07 · Check the circuit with a truth table

For three inputs there are 8 combinations . Work out the OR column first, then use that result with C.

Q = (A ∨ B) ∧ C. Only set Q to 1 when C is 1 and A∨B is 1.

I DO — WORKED EXAMPLE

Follow the worked method

Secure door access

What switches ON?

```
Q = door unlocks
```

Start with the result the question is asking us to control.

What are the alternatives?

```
A ∨ B
```

A card OR PIN works. OR (∨) means at least one; both can also be true. Do not write ⊻ here: XOR would wrongly stop access when both card and PIN are valid.

What must also be true?

```
(A ∨ B) ∧ C
```

The building must be open , so join the options with AND.

What must be OFF?

```
¬D
```

Lockdown must not be active. NOT reverses D.

Join the chunks

```
Q = (A ∨ B) ∧ C ∧ ¬D
```

Work backwards: the final AND checks all requirements. Draw OR, AND and NOT first, then connect them to the final AND.

Use intermediate columns

For Q = (A ∨ B) ∧ ¬C, three inputs give $2^3 = 8$ combinations. Write every input row, calculate A∨B, then ¬C, then combine those two columns with AND.

```
A B C | AvB | ¬C | Q
0 0 0 |  0  |  1 | 0
0 0 1 |  0  |  0 | 0
0 1 0 |  1  |  1 | 1
0 1 1 |  1  |  0 | 0
1 0 0 |  1  |  1 | 1
1 0 1 |  1  |  0 | 0
1 1 0 |  1  |  1 | 1
1 1 1 |  1  |  0 | 0
```

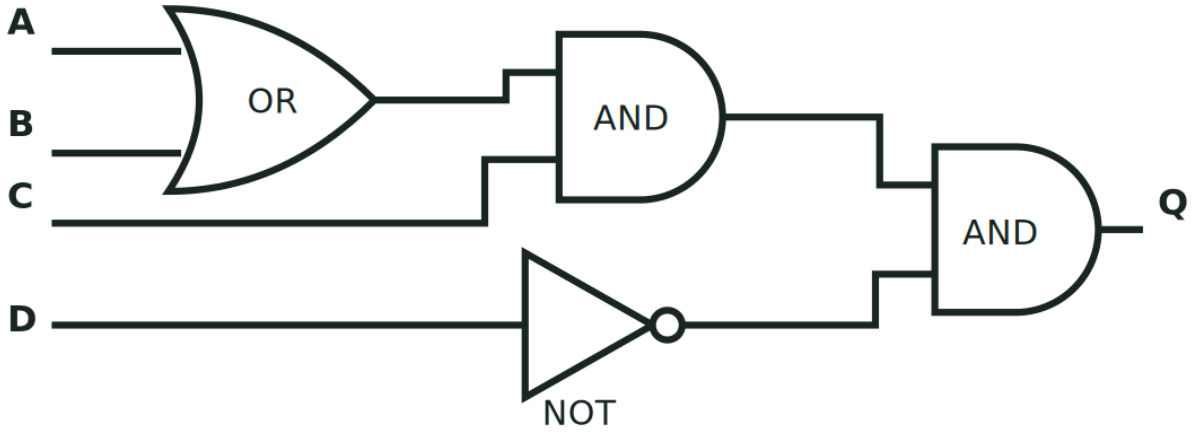
Example row 010: A∨B = 1 and ¬C = 1; therefore Q = 1. Always complete the intermediate columns before Q.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Join the chunks

Work backwards: the final AND checks all requirements. Draw OR, AND and NOT first, then connect them to the final AND.



Secure door circuit: valid card A and correct PIN B feed OR; that output and building open C feed AND; lockdown D feeds NOT; both results enter the final AND to produce Q.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

1. Successful login

A login succeeds only when the user ID is valid AND the password is correct . Find the output, options, requirements and exceptions. Then write the expression and draw the gate.

```
A B | Q
0 0 | ?
0 1 | ?
1 0 | ?
1 1 | ?
```

Method / working:

2. Two test switches

A test lamp lights when exactly one of switches A and B is ON, but not both . Compare this with a different instruction: “at least one switch is on”.

```
A B | EXACTLY ONE | AT LEAST ONE
0 0 |      ?      |      ?
0 1 |      ?      |      ?
1 0 |      ?      |      ?
1 1 |      ?      |      ?
```

Method / working:

3. Library printer

The printer runs if a staff job OR a pupil job is waiting, but only if there is NOT a paper jam . Find the OR group, then the condition that prevents the output.

```
Q = (A ? B) ? ?C
```

Method / working:

4. Delivery bay light

A bay light comes on when movement is detected OR the door is open , but only when power is enabled and the maintenance lock is NOT active. Use all five steps. Decide which gate must feed Q last.

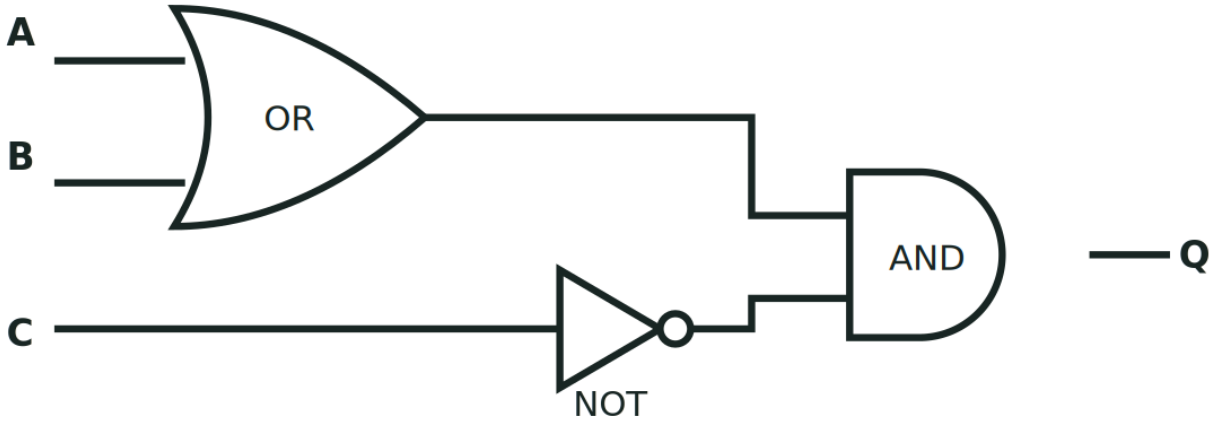
Method / working:

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

OR, NOT, then AND

Two options → OR. Jam absent → NOT C. Both conditions → final AND.



Printer circuit. A and B feed an OR gate. C feeds a NOT gate. The OR and NOT outputs feed a final AND gate to make Q.

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Successful login A login succeeds only when the user ID is valid AND the password is correct . CODE: A = valid user ID B = correct password Q = login succeeds Find the output, options,…

DISCUSS — Compare approaches, walk through one test input, and explain how you would detect or correct a mistake.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

Guided 2: Two test switches A test lamp lights when exactly one of switches A and B is ON, but not both . CODE: A = switch 1 on B = switch 2 on Q = lamp lights Compare this with a different…

DISCUSS — Compare approaches, walk through one test input, and explain how you would detect or correct a mistake.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

Guided 3: Library printer The printer runs if a staff job OR a pupil job is waiting, but only if there is NOT a paper jam . CODE: A = staff job waiting B = pupil job waiting C = paper jam Q =…

DISCUSS — Compare approaches, walk through one test input, and explain how you would detect or correct a mistake.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

Guided 4: Delivery bay light A bay light comes on when movement is detected OR the door is open , but only when power is enabled and the maintenance lock is NOT active. CODE: A = movement detected…

DISCUSS — Compare approaches, walk through one test input, and explain how you would detect or correct a mistake.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Burglar alarm circuit
A burglar alarm has four inputs. Each input is 1 when the condition is true and 0 otherwise:
A: door sensor detects movement
B: motion sensor detects movement
C: alarm is armed
D: test mode is enabled
The alarm sounds (Q = 1) only when at least one sensor detects movement , the alarm is armed , and test mode is off .
Your tasks: (1) Write the Boolean expression for Q. (2) Draw the logic-gate circuit using AND, OR and NOT gates. (3) Label A, B, C, D and Q.

Question 2

Expression and truth table
For P = ¬(A ∨ B) ⊻ C, complete all eight output values.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Create logic from words
A cooling fan runs when temperature is high OR load is high, but only when maintenance mode is NOT enabled. Write the expression and truth table.

Question 4

Expression to gate plan
Convert Q = (A ⊻ B) ∧ ¬C into a gate-by-gate plan and complete its truth table.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A three-input Boolean problem has eight possible input combinations in its truth table.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

AND: both required

CODE:

Q = A ∧ B

Output: login. No OR options or NOT exception. AND joins two requirements.

CODE:

A B | Q

0 0 | 0

0 1 | 0

1 0 | 0

1 1 | 1

IMPROVE: Explain one correction to your own reasoning.

Guided 2 — original CHECK

OR or XOR?

Choose the gate and write Q = A ⊻ B for exactly one . How would the symbol and truth table change for at least one ?

CODE:

A B | EXACTLY ONE | AT LEAST ONE

0 0 | ? | ?

0 1 | ? | ?

1 0 | ? | ?

1 1 | ? | ?

XOR (⊻) versus OR (∨)

Exactly one → XOR: Q = A ⊻ B . At least one → OR: Q = A ∨ B (both allowed).

CODE:

A B | XOR | OR

0 0 | 0 | 0

0 1 | 1 | 1

1 0 | 1 | 1

1 1 | 0 | 1

IMPROVE: Explain one correction to your own reasoning.

Guided 3 — original CHECK

Build two chunks

Write the options in brackets. Apply NOT to the exception . What final gate must join them?

CODE:

Q = (A ? B) ? ?C

Sketch the circuit before revealing the answer.

OR, NOT, then AND

CODE:

Q = (A ∨ B) ∧ ¬C

Two options → OR. Jam absent → NOT C. Both conditions → final AND.

IMPROVE: Explain one correction to your own reasoning.

Guided 4 — original CHECK

Words → gates

Name the output Q.

Bracket the alternative triggers.

Join the power requirement.

Invert maintenance lock.

Write the full expression, then draw.

Can you identify the final gate by working backwards?

Compare each chunk

CODE:

Options (A ∨ B)

Requirement C

Exception ¬D

CODE:

Q = (A ∨ B) ∧ C ∧ ¬D

Draw OR(A,B), AND with C, NOT D, then a final AND . Only reveal once pupils have sketched the circuit.

IMPROVE: Explain one correction to your own reasoning.